IGCSE 0607 Practice Questions — E5.7

Circle theorems II

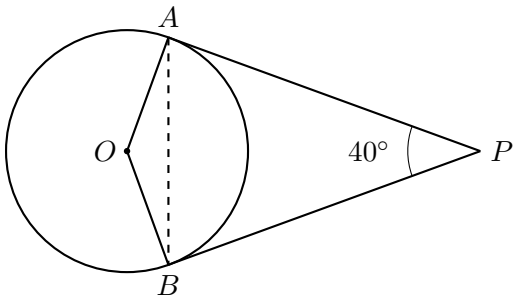
Total Questions: 9 — **Total Marks:** 57 Marks

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1 P is a point outside a circle, centre O.
PA and PB are tangents to the circle at A and B.
Angle $APB = 40°$.



(a) Write down the geometrical reason why triangle PAB is isosceles.

..... **[1]**

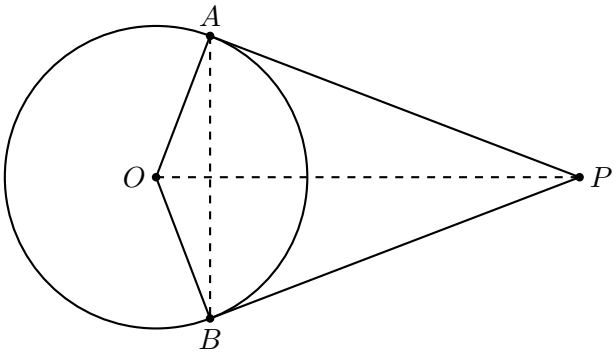
(b) Calculate angle PAB.

Angle $PAB =$ ..... ° **[1]**

(c) Calculate angle AOB.

Angle $AOB =$ ..... ° **[2]**

2 P is an external point.
PA and PB are tangents to a circle, centre O.
The line OP intersects the chord AB at right angles.
$OA = 7$ cm and $PA = 24$ cm.



(a) Write down the length of PB.

..... cm **[1]**

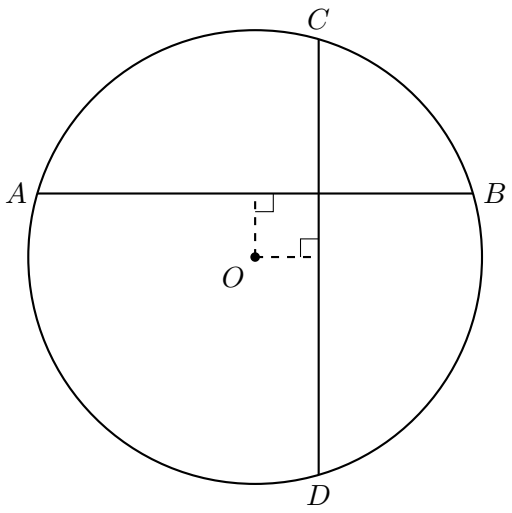
(b) Calculate the length of OP.

..... cm **[2]**

(c) Calculate the shortest distance from O to the chord AB.

..... cm **[3]**

3 The diagram shows a circle, centre O, with radius 25 cm.
AB and CD are two chords of the circle such that $AB = CD$.
The perpendicular distance from O to AB is $(x + 3)$ cm.
The perpendicular distance from O to CD is $(3x - 5)$ cm.



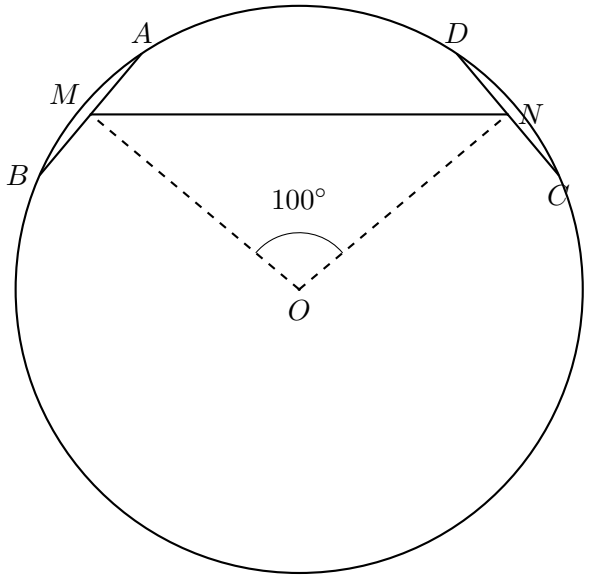
(a) Find the value of x.

$x =$ ..... **[2]**

(b) Calculate the length of the chord AB.

$AB =$ ..... cm **[3]**

4 A, B, C and D are points on a circle, centre O and radius 25 cm.
The chords AB and CD each have a length of 14 cm.
M is the midpoint of AB and N is the midpoint of CD.
Angle $MON = 100°$.



(a) Show that $OM = 24$ cm.

[2]

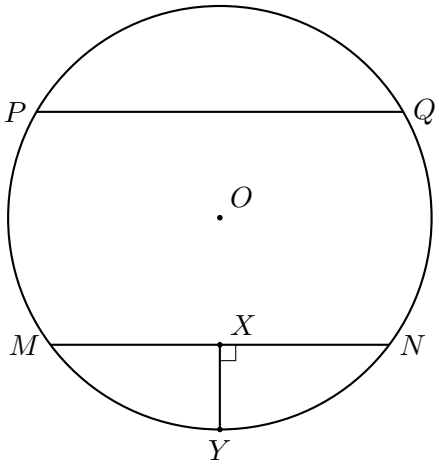
(b) Calculate the length of MN.

..... cm **[3]**

(c) Calculate the area of triangle MON.

..... cm^2 **[2]**

5 A circle has centre O and radius 10 cm.
MN is a chord of length 16 cm.
The perpendicular bisector of MN intersects MN at X and the minor arc MN at Y.
PQ is a second chord which is parallel to MN.
The perpendicular distance between MN and PQ is 11 cm.



(a) Explain why the line containing XY passes through the centre O.

..... **[1]**

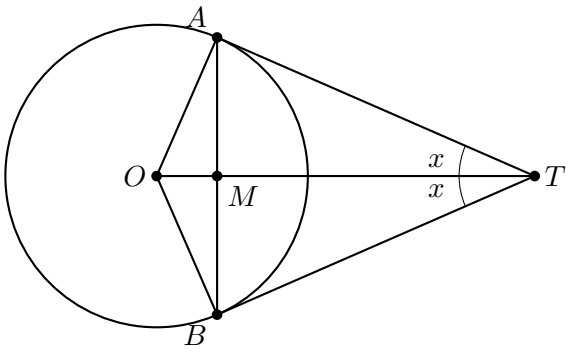
(b) Calculate the length of XY.

$XY =$ ..... cm **[3]**

(c) Calculate the length of the chord PQ.

$PQ =$ ..... cm **[4]**

6 TA and TB are tangents to a circle, centre O and radius r, at the points A and B.
The line TO intersects the chord AB at M.
Angle ATO = Angle $BTO = x$.



(a) Show that the length of the chord AB is $2r\cos x$.

[2]

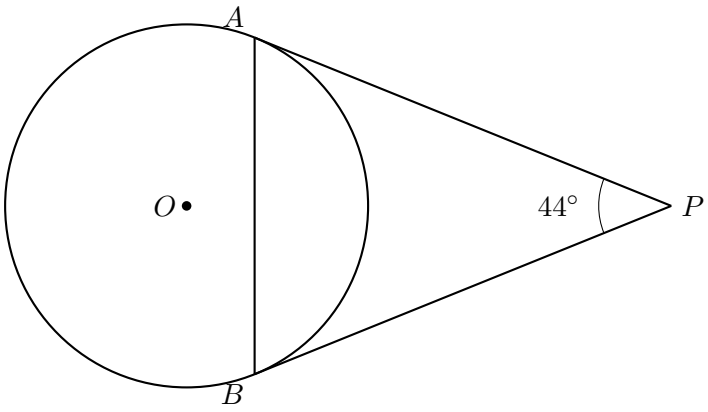
(b) A second chord BC is drawn such that $BC = AB$.
Write down the shortest distance from O to the chord BC, in terms of r and x.

..... **[1]**

(c) The area of triangle OAB is equal to the area of triangle ATB.
Find the exact value of x.

$x =$ ..... $^\circ$ **[5]**

7 The diagram shows a circle with centre O.
P is an external point.
The lines PA and PB are tangents to the circle at the points A and B.
Angle $APB = 44°$.



NOT TO SCALE

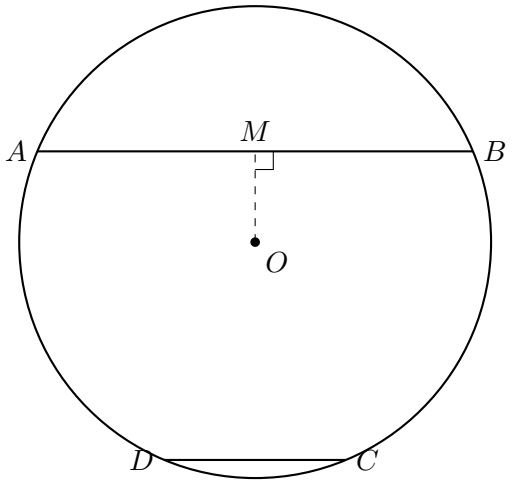
(a) Calculate angle PAB.

Angle $PAB =$ ..... ° **[2]**

(b) Calculate angle AOB.

Angle $AOB =$ ..... ° **[2]**

8 The diagram shows a circle, centre O, with radius 13 cm.
AB is a chord of the circle of length 24 cm.
The point M is the midpoint of the chord AB.
Another chord CD is drawn inside the circle.



NOT TO SCALE

(a) Calculate the perpendicular distance, OM, from the centre of the circle to the chord AB.

$OM =$ ..... cm **[3]**

(b) The perpendicular distance from the centre O to the chord CD is also 5 cm.
Write down the length of the chord CD.

$CD =$ ..... cm **[1]**

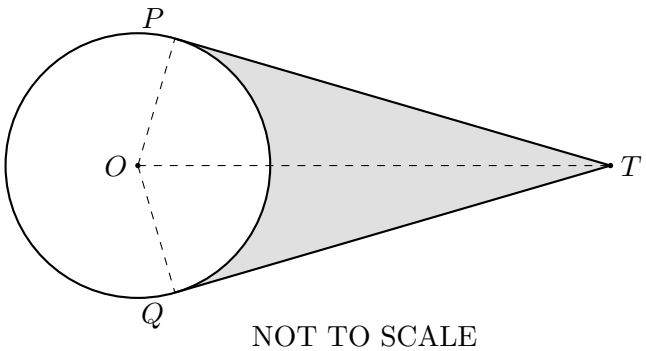
(c) Calculate the area of triangle OAB.

Area =

.....

cm^2 **[2]**

9 The diagram shows a circle with centre O and radius 7 cm.
The lines TP and TQ are tangents to the circle at the points P and Q.
The distance from the centre O to the external point T is 25 cm.



NOT TO SCALE

(a) Show that the length of the tangent TP is 24 cm.

[2]

(b) Calculate angle PTQ.

Angle PTQ = ... ° **[3]**

(c) Calculate the area of the shaded region bounded by the tangents TP, TQ and the minor arc PQ.

Area =

.....

cm^2 **[4]**